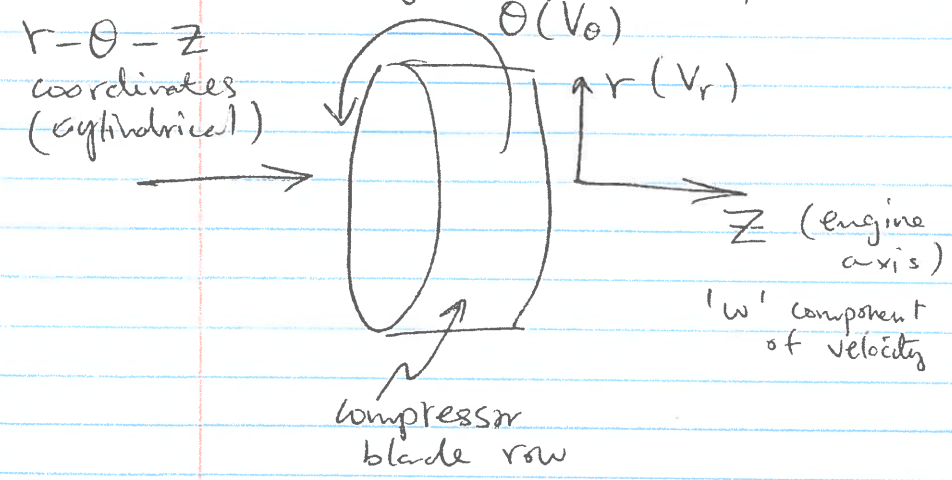
TURBOMACHINERY AERODYNAMICS

* Inlet Guide Vane (IGV): prepares angle of the flow entering the first stage of the compressor gets it ready for first stage rotor blade.

rotor, stator

Fluid moving thru a compressor experiences a torque from the moving rotor blades (in the θ-direction), i.e. a V_θ, component of velocity, is imparted to the flow.

$$V_\theta = r\dot{\theta} = r\omega$$ where ω is angular velocity in radians/sec.



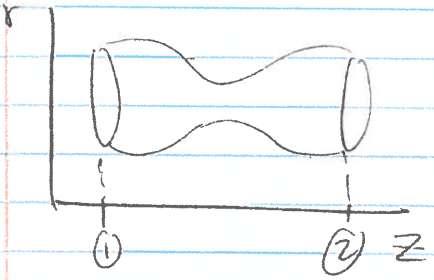
rotor rotating at $\omega \left(= \frac{d\theta}{dt}\right)$

V_θ = "swirl" velocity component in θ direction

Torque = $rF_\theta = T$

(F_θ = local force in θ direction at a r location from hub)

It can be shown using the moment of momentum theorem based on Newton's law that the torque 'T' on the fluid in a stream tube provided by the rotor blades between station 1 & 2 in the z-direction



$$T = \dot{m}\left[(rV_\theta)_2 - (rV_\theta)_1\right]$$

$\dot{m}$ = $\dot{m}$ of air in a defined streamtube passing thru a compressor blade row

'rV_θ' is angular momentum at a given station

Furthermore, power considerations demand that the mechanical power between (1) & (2)

$F_\theta \cdot \frac{(r d\theta)}{dt}$ ⟵ $\boxed{\dot{W} = \text{Power}_{1\to 2} = \omega T = \dot{m}\omega\left[(rV_\theta)_2 - (rV_\theta)_1\right]}$

but we also know the energy equation:

$\dot{m}\left(h_{t_2} - h_{t_1}\right) = \dot{W}_{\text{adiabatic compressor}}$: $h_t = c_p T_t$

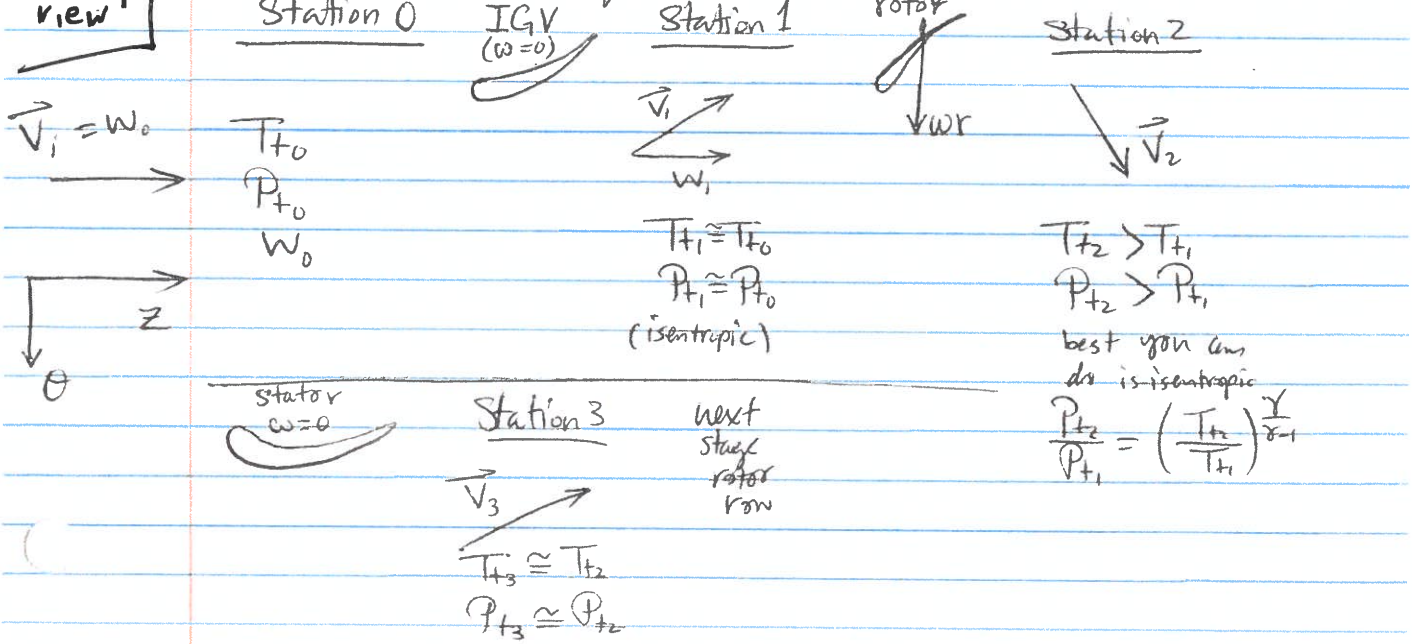
$c_p\left(T_{t_2} - T_{t_1}\right) = h_{t_2} - h_{t_1} = \frac{\omega T}{\dot{m}}$

or $\boxed{c_p\left(T_{t_2} - T_{t_1}\right) = h_{t_2} - h_{t_1} = \omega\left[(rV_\theta)_2 - (rV_\theta)_1\right]}$ *

- if blade is not moving (stator), $\omega = 0 \Rightarrow$ no work
- only rotor provides work.

'cascade view'

'unwrap' the compressor at a given r location:



If the process is isentropic thru rotor

$$** \quad \frac{P_{t2}}{P_{t1}} = \left(\frac{T_{t2}}{T_{t1}}\right)^{\frac{\gamma}{\gamma-1}}$$

if NOT isentropic, use loss coefficient
i.e. $\frac{P_{t2}}{P_{t1}} = \left(\frac{T_{t2}}{T_{t1}}\right)^{\frac{\gamma}{\gamma-1}\eta_c}$ etc...

combine / use * & ** with assumed small Mach numb ($\rho \sim \rho_{STDP}$) (≤ 0.5) so ρ does not change much

→ you can show algebraically for small turning

NON-dimensional ΔP_t thru rotor row

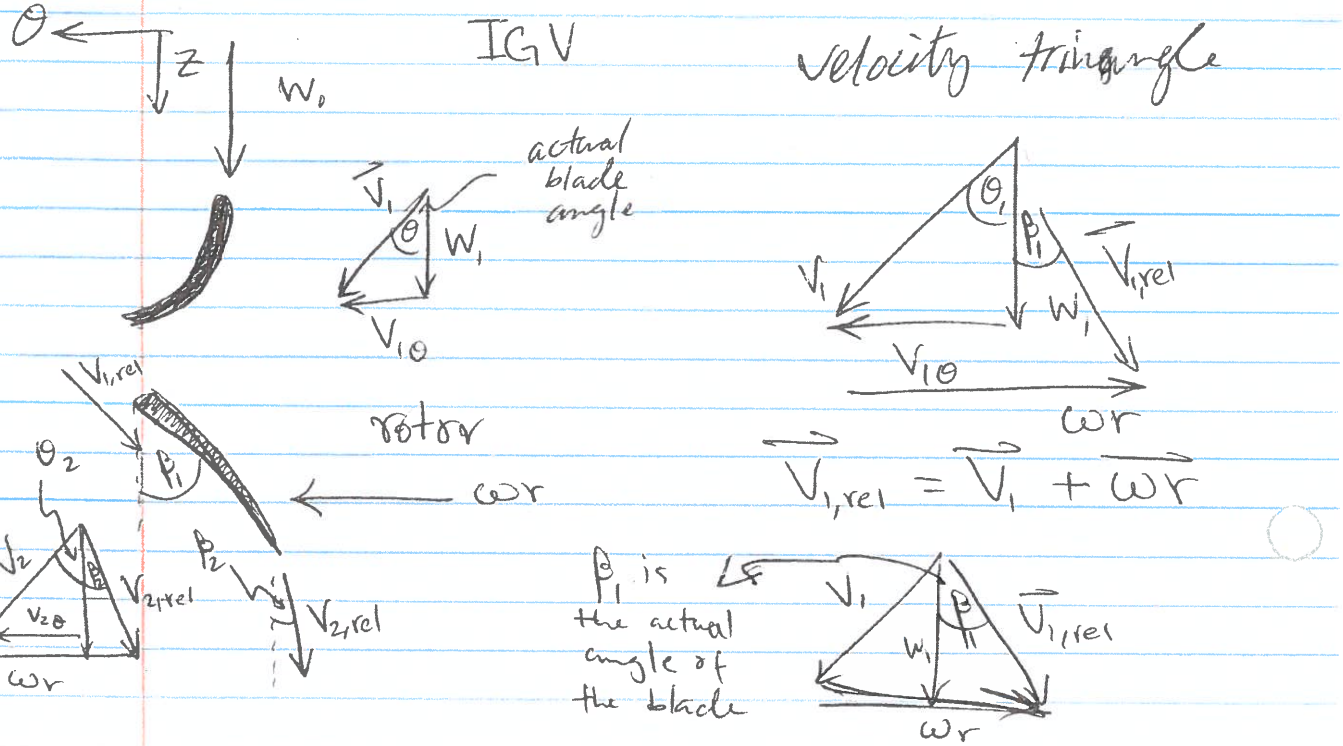
$$\frac{P_{t2} - P_{t1}}{\rho_0 W_0^2} \cong \frac{\omega}{W_0^2}\left[(rV_\theta)_2 - (rV_\theta)_1\right]$$

expression for P_t rise in terms of change in angular momentum

example: if $\Pi_c = 20$ and 13 stages (F-15 F-100 engine) (for $P_{t0} = 101325$ N/m²)

ΔP_t single stage $\cong 53{,}000$ N/m²

→ It would be highly useful to define the local flow turning in terms of actual (physical) local blade angles of IGV, rotor, stator blades.

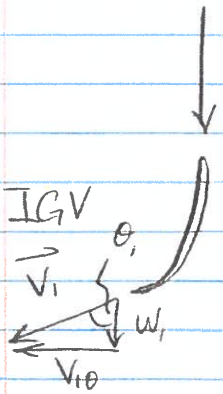


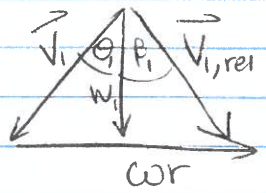
'Repeat'

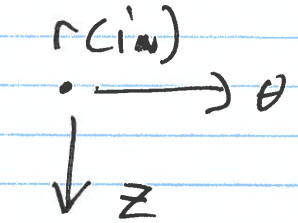
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$$\frac{P_{t_2} - P_{t_1}}{\rho_0 W_0^2} = \frac{\omega}{W_0^2}\left[(rV_\theta)_2 - (rV_\theta)_1\right]$$

0

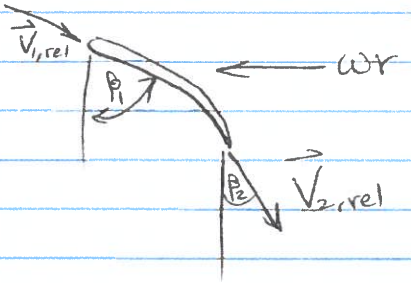






'velocity triangle'

1

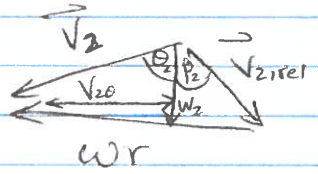


Relative → 'rel' to rotor

in rotor frame → physical

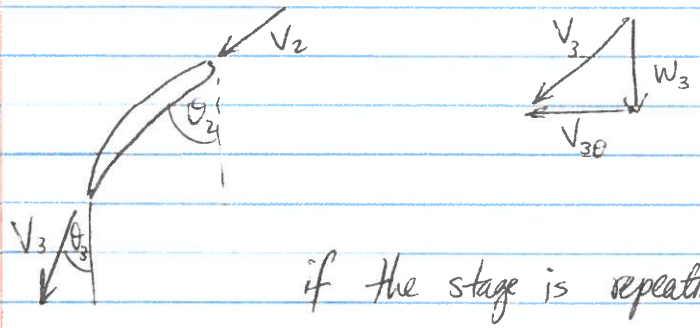
β's are actual blade angles (leading edge and trailing edge)

2



→ exit's velocity triangle

stator



θ_2 & θ_3 are actual (physical) LE & TE angles of stator!

if the stage is repeating: $\vec{V}_3 = \vec{V}_1$

3

$\theta_3 = \theta_1$

assume $W_0 = W_1 = W_2 = W_3$ and r doesn't change much thru a stage, ρ does not change (M<.5

Statement: velocity triangles relate engine frame/rotor relative velocities & angles!

Observe these Relationships from velocity triangles:

$$\omega r - V_{1\theta} = W_\theta \tan\beta_1$$

$$\omega r - V_{2\theta} = W_\theta \tan\beta_2$$

$$\Rightarrow V_{2\theta} - V_{1\theta} = W_\theta (\tan\beta_1 - \tan\beta_2)$$

since $\frac{V_{2\theta}}{W_\theta} = \tan\theta_2$; $\frac{V_{1\theta}}{W_\theta} = \tan\theta_1$

$$\tan\beta_1 = \frac{\omega r}{W_\theta} - \tan\theta_1$$

$$\tan\beta_2 = \frac{\omega r}{W_\theta} - \tan\theta_2$$

} relates β's and θ's!!

from * then →

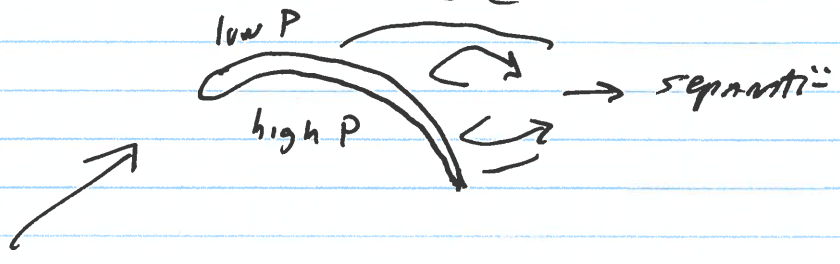
$$\frac{P_{t_2} - P_{t_1}}{\rho_\theta W_\theta^2} = \frac{\omega r}{W_\theta}(\tan\beta_1 - \tan\beta_2)$$

→ ΔP_t for a stage in terms of physical LE/TE blade angles of rotor!!

approximate! based on incompressible approx. and small flow turning!

* To increase the ΔP_t across the stage:
  - increase ω, W_θ but limited by blade tip effects
  - Increase amount of turning thru the rotor; $(\tan\beta_1 - \tan\beta_2)$, however, limited by seperation due to an adverse pressure gradient. The static pressure, P, goes up, seperation may occur, due to too much blade curvature.

Too much blade turning (curvature):



* 2 Engineering coefficients are generally used in the business for assessing seperation risk. based on empirical data.

1) Degree of Reaction, $R = \frac{\Delta P_{rotor}}{\Delta P_{stage}}$ "or defined about a stage"

$R = 1 - \frac{\Delta P_{static,\,stator}}{\Delta P_{stage}}$ $\therefore \Delta P_{stage} = \Delta P_{rotor} + \Delta P_{stator}$

P → static pressure

Driver representation phenomena: P.G. $= \frac{dP}{dx}$ pressure gradient

'spends' P.G. between rotor & stator

$R \sim 0.5$ is good (equal blade 'loading')

R can be shown to be:

$R_{stage} = 1 - \frac{(V_{\theta_2} + V_{\theta_1})}{2\omega r} = \frac{w(\tan\beta_2 + \tan\beta_1)}{2\omega r}$ (show using velocity triangle info)

2) Diffusion Factor, D (aka "blade loading factor")

describes both the pressure gradient effect and blade curvature effect on the pressure gradient.

(hence we have D_{stator}, D_{rotor})

$D = \left(1 - \frac{|\vec{V}_e|}{|\vec{V}_i|}\right) + \frac{|\Delta V_\theta|}{2\sigma|\vec{V}_i|}$: i & e denote inlet and exit for a given blade row (stator or rotor)

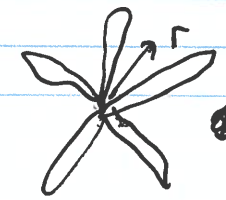
pressure gradient effect — blade curvature effect

$D \leq 0.6$ is 'good' (small D is 'good')

σ = 'blade solidity' $= \frac{\text{chord}}{\text{spacing}} = \frac{c}{s}$

at a given r location

- it changes with r



• s grows with r...

* It can be shown: (from velocity triangles)

$$D_{rotor} = 1 - \frac{\cos\beta_1}{\cos\beta_2} + \frac{1}{2\sigma}\left|(\tan\beta_1 - \tan\beta_2)\right| \frac{1}{\cos\beta_1}$$

$$D_{stator} = 1 - \frac{\cos\theta_2}{\cos\theta_3} + \frac{\left|\tan\theta_3 - \tan\theta_2\right|}{2\sigma \sec\theta_2}$$

This then gives a means to 'assess' whether a compressor blade design is acceptable at a given r

---

So... * How does one begin to design a compressor? (i.e., choose β's, θ's across the r range; hub to tip).

'Free Vortex' compressor stage 'base line'

we KNOW $\frac{\Delta P_t}{\rho_0 W_0^2} \cong \frac{\omega r}{W_0}(\tan\beta_1 - \tan\beta_2)$

(desirable to have)

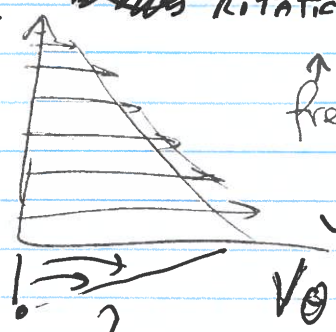
* we want ΔP_t as constant as possible from hub to tip (across r); don't like gradients, at a given axial location ('z')

↳ true for 'Free Vortex' compressor stage.

obviously, P_t going up as progress thru compressor axially

(rV_θ) = const. with r → maintains constant P_t with r

~~Rotational~~ Core' ~~Flow~~)



↳ Rotational 'core'

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Free Vortex Compressor "A place to start"

- blades are designed to ensure $(rV)_\theta$ = const.

ensures $P_t(r)$ = const. at a z-position

From continuity $u_r = 0$ (radial velocity component) = 0

∴ no radial movement of the streamtubes.

Example:

$\rho = 1.225\ kg/m^3$, RPM = 10,000

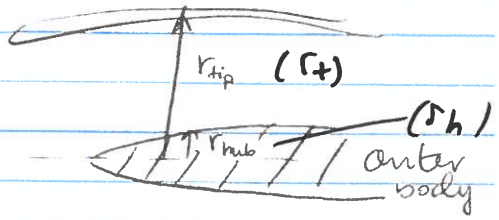
So ∴ $\omega = 1047.2$ rad/s

$V_\theta = 100$ m/s

require $V_{\theta_1} = 10$ m/s at the hub $= V_{\theta_{1,h}}$

1 → 2 rotor

2 → 3 stator



let $r_{hub} = 0.4$m (hub radius)

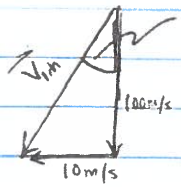
$r_{tip} = 0.56$m (tip radius)

require $\Delta P_t = 50{,}000$ Pa (single stage)

hence $(\omega r)_h = 418.7$ m/s, $(\omega r)_t = 576.0$ m/s

A. at hub:

At exit of the upstream ↑ stator

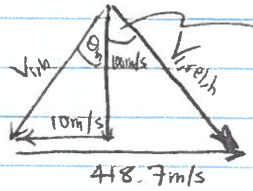


$\theta_{1,hub} = 5.71° = \arctan\left(\frac{10}{100}\right)$

∴ $V_{1,h} = 100.5$ m/s

B. at hub:

entering the rotor



$\beta_{1,h} = 76.3° = \arctan\left(\frac{408.7}{100}\right)$

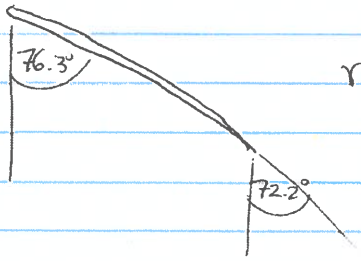
∴ $V_{t,rel,h} = 420.8$ m/s

(a)

we know $\frac{P_{t2} - P_{t1}}{\rho W_0^2} = \frac{\omega r}{W_0}(\tan\beta_1 - \tan\beta_2)$

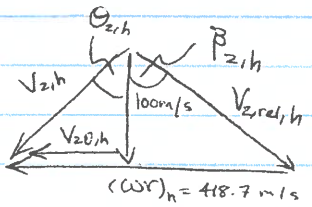
so $\therefore \beta_2 = 72.2°$ (for $\Delta P_t = 50\ kPa$)

rotor sketch at hub



relatively flat

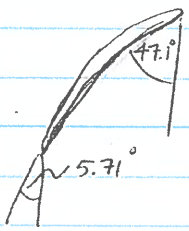
C. at hub: exiting the rotor



$\therefore V_{2,rel,h} = 327\ m/s$

$V_{\theta,2h} = 107.6\ m/s$
$\theta_{2,h} = 47.1°$
$V_{2,h} = 147\ m/s$

stator sketch at hub (for repeating stage)



$\therefore \theta_{3h} = \theta_{1h}$ repeating

$^0R_h = \frac{\Delta P_{rotor}}{\Delta P_{stage}} = 0.86$ (rotor a bit 'loaded' in comparison to the stator)

- the flow over the rotor is more likely to seperate.

from before

$$^0R_h = \left\{\frac{W_0(\tan\beta_2 + \tan\theta_1)}{2\omega r}\right\}_{hub} = 0.86$$

D. Repeat at tip where $(\omega r)_t = 576$ m/s

(since $rV_\theta = \text{const.}$, so $r_h V_{\theta,h} = r_t V_{\theta,t}$)

$\therefore V_{\theta_1,t} = 7.3$ m/s

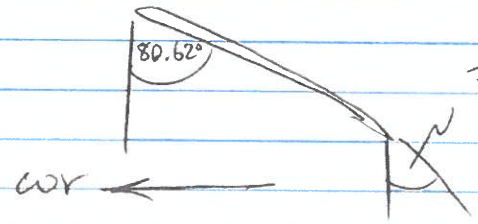
at (tip): $\theta_{1,t} = 4.16°$

(work flow analysis) $\beta_{1,t} = 80.62°$

$\beta_{2,t} = 78.64°$

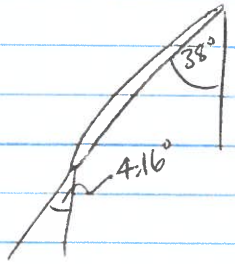
$\theta_{2,t} = 38.0°$

rotor sketch at tip



Virtually flat-!

stator sketch at tip



And $^{0}R_{tip} = 0.925$ $\left(= \frac{\Delta P_{Rotor}}{\Delta P_{Stator}}\right)$

Rotor is 'loaded' with more static pressure rise than stator

So rotor and stator geometry are 'rotating' clockwise from tip to hub. (with ωr direction as shown)

Analysis of free-vortex compressor stage: Method

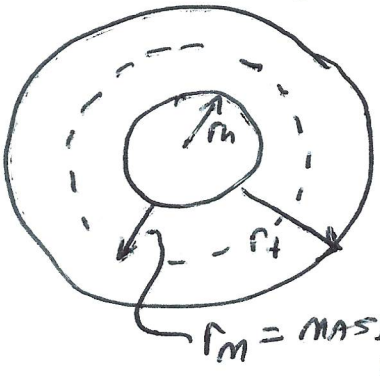
For a given stage $^{o}R = 1 - \frac{(V_{\theta 2} + V_{\theta 1})}{2\omega r}$

(degree of reaction)

For a 'free-vortex' design: Across a plane $rV_\theta = \underline{\text{const}}$

so $^{o}R = 1 - \frac{\text{constant}}{2\omega r^2}$ { as long as the 'r' of a streamtube does not change much across a single stage ...

Now, assume that the value of ^{o}R at the 'mass-averaged' radius is specified. The mass-averaged radius is that radius at which mass flow rate of air, $\dot{m}$, is evenly split between r_t (tip radius) and r_h (hub radius). For uniform axial velocity and density of gas not changing with r, $\dot{m}$ is proportional to AREA



hence $\pi r_t^2 - \pi r_m^2 = \pi r_m^2 - \pi r_h^2$

or $r_m = \sqrt{\frac{r_t^2 + r_h^2}{2}}$

[Define $^{o}R_m = {}^{o}R$ at r_m & Note it would be good to have $^{o}R_m \sim 0.5$ at r_m ! (so rotor & stator blade rows at r_m would be 'equally loaded' in terms of static pressure rise ...]

Hence $^{o}R_m = 1 - \frac{const}{2\omega r_m^2}$ or $const = 2\omega r_m^2 (1 - {}^{o}R_m)$

So, in general: $^{o}R = 1 - \frac{r_m^2}{r^2}(1 - {}^{o}R_m)$

on substituting in the r_m expression

$$^{o}R = 1 - \frac{(r_t^2 + r_h^2)}{r^2}\left(\frac{1 - {}^{o}R_m}{2}\right)$$

Hence, at hub

$$^{o}R_h = 1 - \left[\left(\frac{r_t}{r_h}\right)^2 + 1\right]\left[\frac{1 - {}^{o}R_m}{2}\right] \quad *$$

Σ since $^{o}R = 1 - \frac{(V_{\theta_2} + V_{\theta_1})}{2\omega r}$ (shown when ^{o}R defined, previous notes)

$$\frac{V_{\theta_2 h} + V_{\theta_1 h}}{w_0} = (1 - {}^{o}R_h)\frac{2\omega r_h}{w_0} \quad (A)$$

we know (showed):

$$\frac{\Delta P_t}{\rho w_0^2} = \frac{\omega}{w_0^2}\left[(rV_\theta)_2 - (rV_\theta)_1\right]$$

thru single stage!

so: $$\frac{V_{\theta_2 h}}{w_0} - \frac{V_{\theta_1 h}}{w_0} = \left[\frac{\Delta P_t}{\rho w_0^2}\right]\frac{w_0}{\omega r_h} \quad (B)$$

From (A) & (B) solve for $\frac{V_{\theta 2h}}{W_0}$ & $\frac{V_{\theta,h}}{W_0}$

obtain:

$$\left\{ \begin{aligned} \frac{V_{\theta 2h}}{W_0} &= \frac{1}{2}\left[\frac{2\omega r_h}{W_0}\left(1-{}^{o}R_h\right)+\frac{1}{\left(\frac{\omega r_h}{W_0}\right)}\left(\frac{\Delta P_t}{\rho_0 W_0^2}\right)\right] \\ \frac{V_{\theta 1h}}{W_0} &= \frac{1}{2}\left[\frac{2\omega r_h}{W_0}\left(1+{}^{o}R_h\right)-\frac{1}{\left(\frac{\omega r_h}{W_0}\right)}\left(\frac{\Delta P_t}{\rho_0 W_0^2}\right)\right] \end{aligned} \right\}$$

(where ${}^{o}R_h$ is a function of ${}^{o}R_m, \frac{r_t}{r_h}$ → see * last page)

So we have obtained θ (swirl) velocities at the hub as a function of

$$\frac{\omega r_h}{W_0}, \frac{\Delta P_t}{\rho_0 W_0^2}, \frac{r_t}{r_h}, {}^{o}R_m$$

But since rV_θ = CONSTANT ACROSS AN AXIAL PLANE for free-vortex compressor ...

$$rV_{\theta 2} = r_h V_{\theta 2h} \quad \& \quad rV_{\theta_1} = r_h V_{\theta,h}$$

SO ... for ANY Radius

$$\frac{V_{\theta 2}}{W_0} = \left(\frac{r_h}{r}\right)\left(\frac{V_{\theta 2h}}{W_0}\right) \quad \& \quad \frac{V_{1\theta}}{W_0} = \left(\frac{r_h}{r}\right)\left(\frac{V_{\theta 1h}}{W_0}\right)$$

hence $\left\{ \frac{V_{\theta 1}}{W_0} \;\&\; \frac{V_{\theta 2}}{W_0} \right\}_{\text{at ANY } r} \Rightarrow f\left(\frac{r_h}{r}, \frac{\omega r_h}{W_0}, \frac{\Delta P_t}{\rho W_0^2}, \frac{r_t}{r_h}, {}^{\circ}R_m\right)$

But this in turn defines the blade angles

$\underbrace{\theta_1, \theta_2}_{\text{stator}}$ & $\underbrace{\beta_1, \beta_2}_{\text{rotor}}$ as functions of $\left(\frac{r_h}{r}, \frac{\omega r_h}{W_0}, \frac{\Delta P_t}{\rho W_0^2}, \frac{r_t}{r_h}, {}^{\circ}R_m\right)$

showed previously → since we know from velocity triangles that

$$\tan\theta_1 = \frac{V_{\theta 1}}{W_0}, \quad \tan\beta_1 = \frac{\omega r}{W_0} - \frac{V_{\theta 1}}{W_0}$$

$$\tan\theta_2 = \frac{V_{\theta 2}}{W_0}, \quad \tan\beta_2 = \frac{\omega r}{W_0} - \frac{V_{\theta 2}}{W_0}$$

& since ${}^{\circ}R$ and D_{Rotor} & D_{Stator} are defined in terms of blade angle, (see notes), these are then defined as well !

Recall:

$$^{o}R = \frac{\tan\beta_2 + \tan\beta_1}{2\frac{\omega r}{W_o}} = \frac{(\tan\beta_2 + \tan\beta_1)}{2\frac{\omega r_h}{W_o}}\left(\frac{r_h}{r}\right)$$

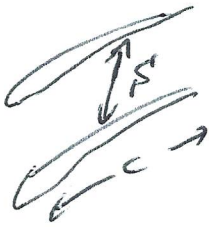
$$D_{Rotor} = 1 - \frac{\cos\beta_1}{\cos\beta_2} + \frac{1}{2\sigma}\left|(\tan\beta_1 - \tan\beta_2)\right|\left(\frac{1}{\cos\beta_1}\right)$$

ξ

$$D_{Stator} = 1 - \frac{\cos\theta_2}{\cos\theta_3} + \frac{|\tan\theta_3 - \tan\theta_2|}{2\sigma\sec\theta_2}$$

(Recall for repeating stage $\rightarrow \theta_3 = \theta_1$)

Note that σ ('solidity') is a function of r



$$\sigma = \frac{c}{s} = \frac{c}{\alpha r}$$

& assume σ_m (solidity at mass-averaged radius) is given ...

$$\sigma_m = \frac{c}{\alpha r_m}$$

or $c = \sigma_m \alpha r_m$ or $\alpha = \frac{c}{\sigma_m r_m}$ so ...

$$\boxed{\sigma = \left(\frac{r_m}{r}\right)\sigma_m}$$

but $\frac{r_m}{r} = \sqrt{\frac{\frac{r_t^2}{r^2} + \frac{r_h^2}{r^2}}{2}} = \sqrt{\frac{\left(\frac{r_h}{r}\right)^2\left[\left(\frac{r_t}{r_h}\right)^2 + 1\right]}{2}}$

so ...

$$\sigma = \sigma_m \sqrt{\frac{\left(\frac{r_h}{r}\right)^2\left[\left(\frac{r_t}{r_h}\right)^2+1\right]}{2}}$$

So for a free vortex compressor stage, you can find $\theta_1 (=\theta_3)$, θ_2, β_1, β_2, ${}^0R$, D_{Rotor}, D_{Stator} at any radius between r_h & r_t

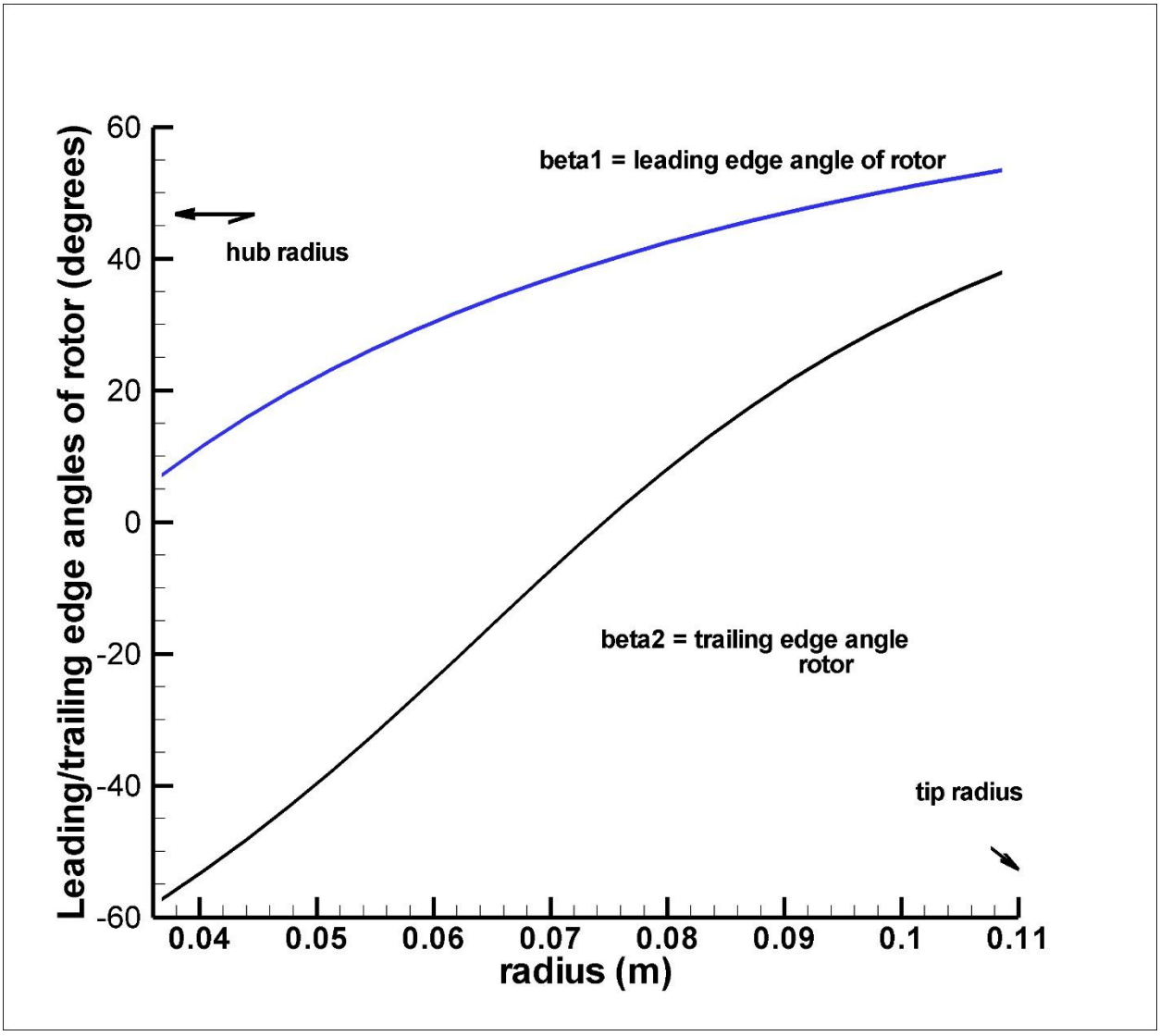
Knowing $\frac{r_h}{r}$, $\frac{r_t}{r_h}$, $\frac{\omega r_h}{w_0}$, $\frac{\Delta P_t}{\rho w_0^2}$, ${}^0R_m$, σ_m

0.5 is 'good'!

Example

Free-Vortex Stage Design: ΔP_t (stage) = 25881 N/m^2 W_0 = 154.15 m/s ^oR_m = 0.5 ρ_0 = 1.2978 kg/m^3

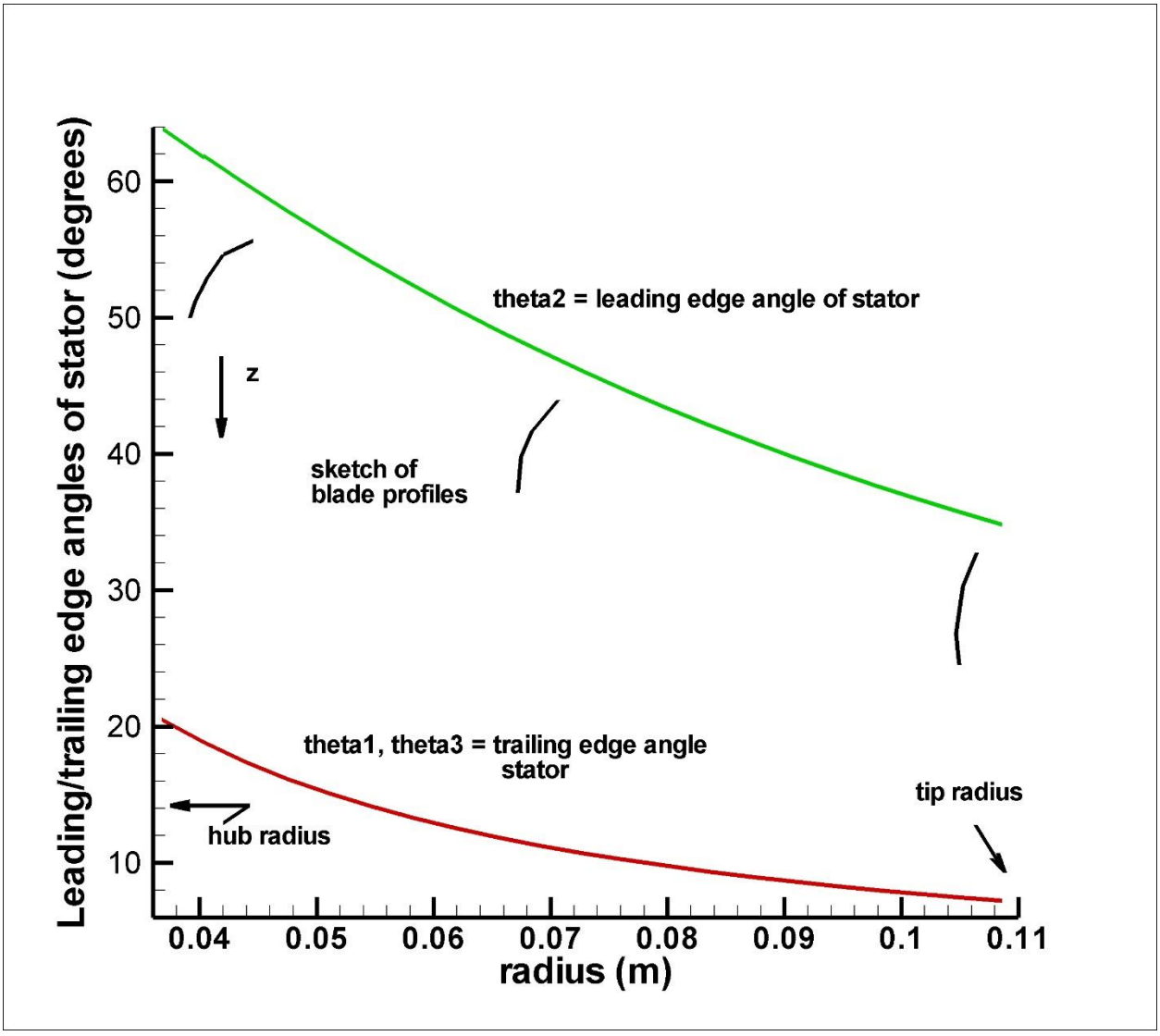
σm = 1.0 rt = 0.1086 m rh = 0.03676 m ω = 2094.4 rad/sec (20000 RPM)



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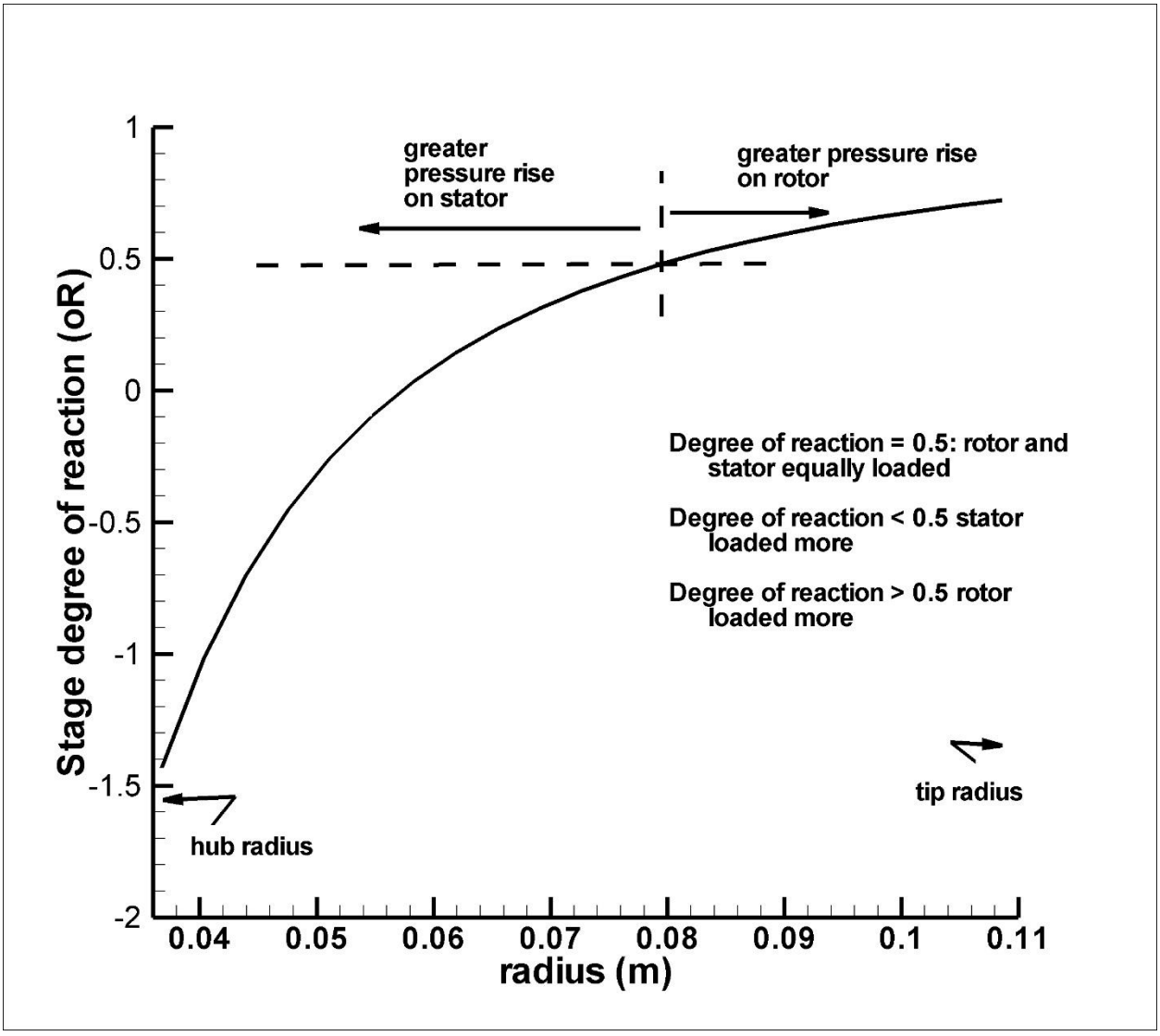
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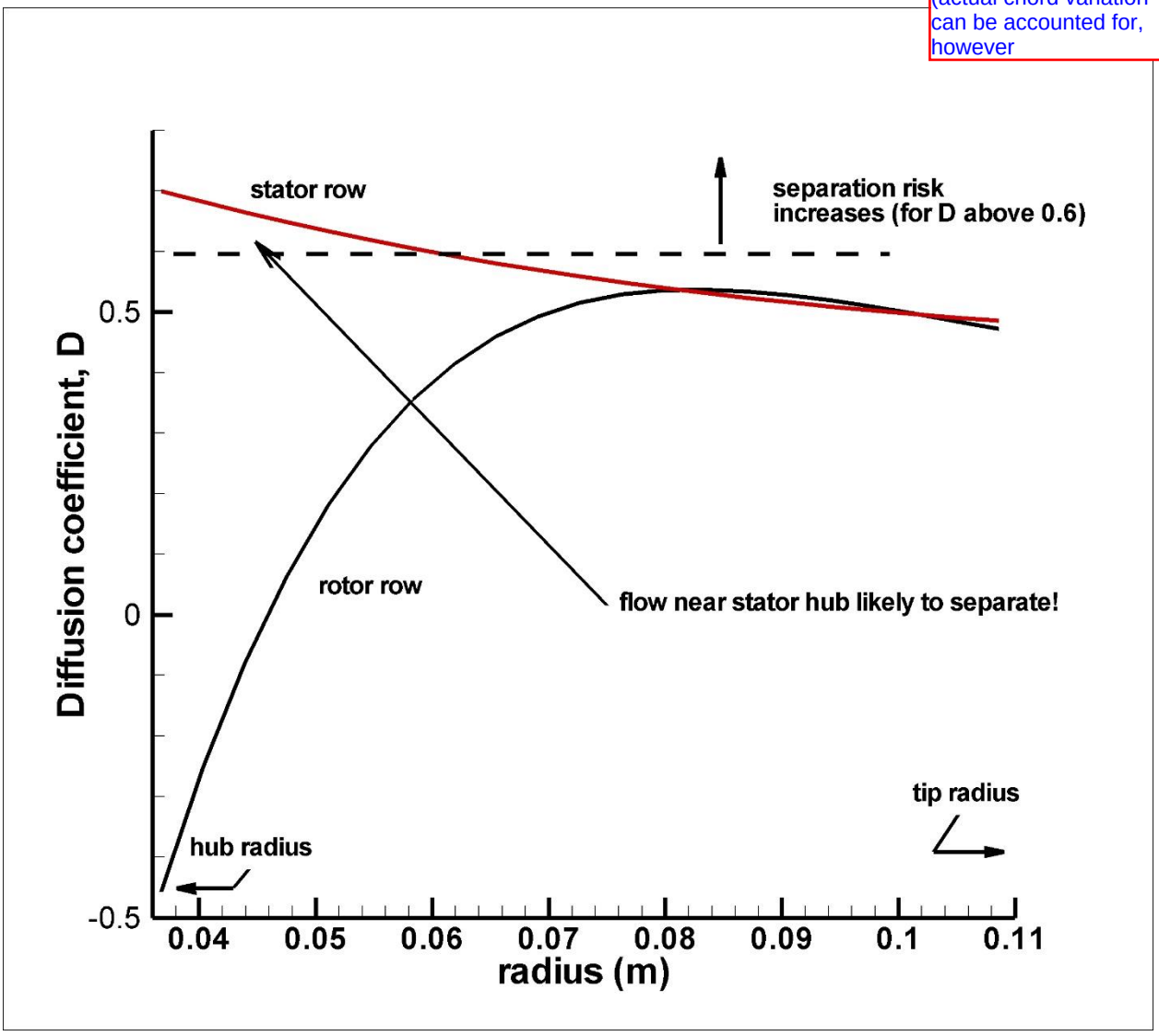


Example

Free-Vortex Stage Design: ΔP_t (stage) = 25881 N/m^2 W_0 = 154.15 m/s ^oR_m = 0.5 ρ_0 = 1.2978 kg/m^3

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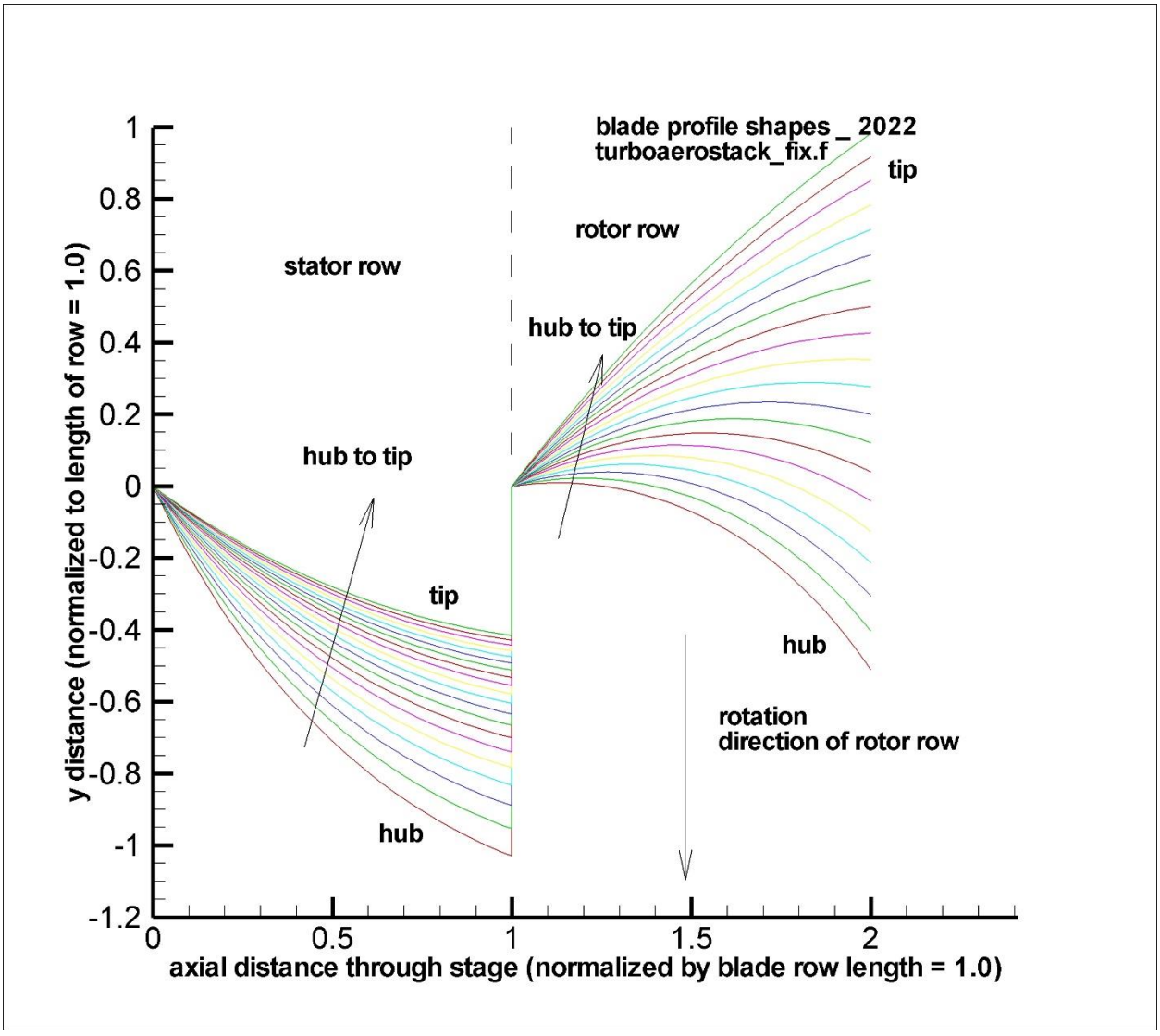
assumes a constant chord for Diffusion factor (load factor) calculation (actual chord variation can be accounted for, however



Example

Free-Vortex Stage Design: ΔP_t (stage) = 25881 N/m^2 W_0 = 154.15 m/s ^oR_m = 0.5 ρ_0 = 1.2978 kg/m^3

σm = 1.0 rt = 0.1086 m rh = 0.03676 m ω = 2094.4 rad/sec (20000 RPM)





Example: pretty representitive numbers

$\sigma = \frac{c}{s}$

Given $\frac{\Delta P_t}{\rho_0 W_0^2} = 0.9$, $\frac{r_t}{r_h} = 2.646$

$^0R_m = 0.5$, $\sigma_m = 1.0$ (typical value for gas turb

remember: 'm' denotes mass-averaged radius

The number of blades determines σ really

$D \leq 0.6$ is good

Consider two values of blade speed $\frac{\omega r_h}{W_0} = 0.5, 0.7$

$\frac{\omega r_h}{W_0}$	$D_{\text{rotor hub}}$	$D_{\text{rotor tip}}$	$D_{\text{stator hub}}$	$D_{\text{stator tip}}$	$^0R_{\text{tip}}$	$^0R_{\text{hub}}$
0.5	-0.179	0.56	0.74 (problem)	0.55	5/7	-1.0
0.7	-0.35	0.38	0.59	0.43	—	— (weren't calculated)

$D_{\text{stator hub}}$ is too high for $\frac{\omega r_h}{W_0} = 0.5$, so put more 'loading' on the rotor by adjusting the blade angles near the hub, etc.

* You can lower D by increasing $\frac{\omega r_h}{W_0}$ to some extent (blade tip effects), decrease $\frac{r_t}{r_h}$ (reduces $\dot{m}$, though) reduce ΔP_t (more stage for desired π_c).

So, free vortex machine is just a 'place to start'.

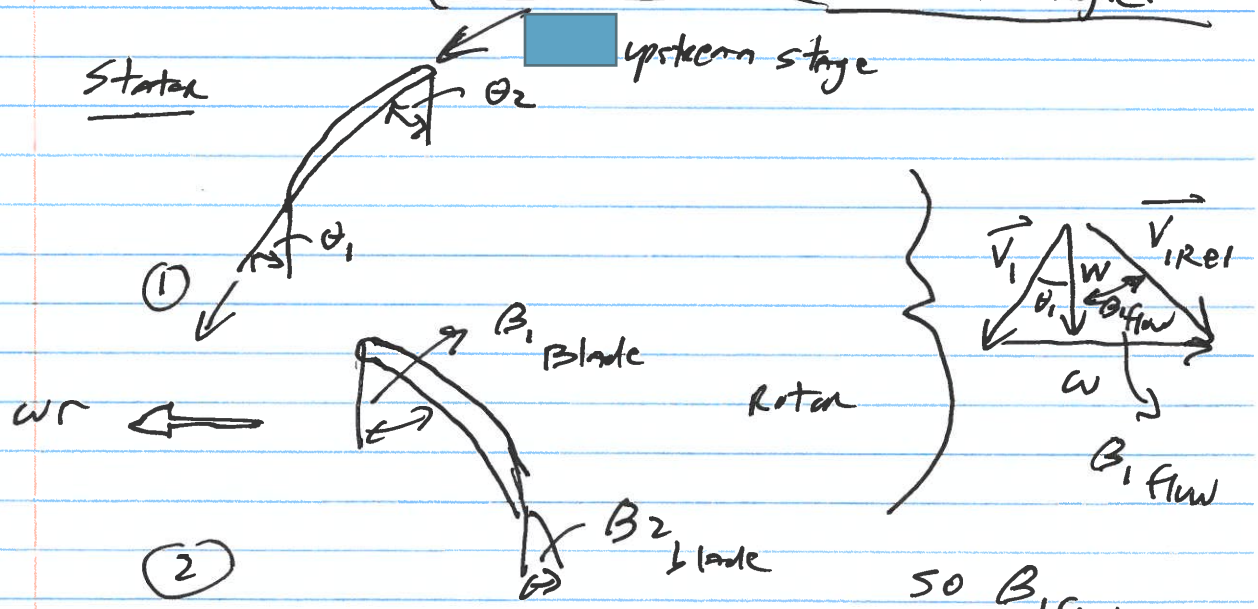
R, D change too dramatically with r.

The best thing to do is to modify the design to allow r variation in streamtube path especially at hub and radius! ($U_r \neq 0$) (helps R & D)

You can always change the stator angles during startup to avoid unwanted phenomena (constant stall, windmilling, etc.).

Compressor Starting (Axial velocity thru compressor small)

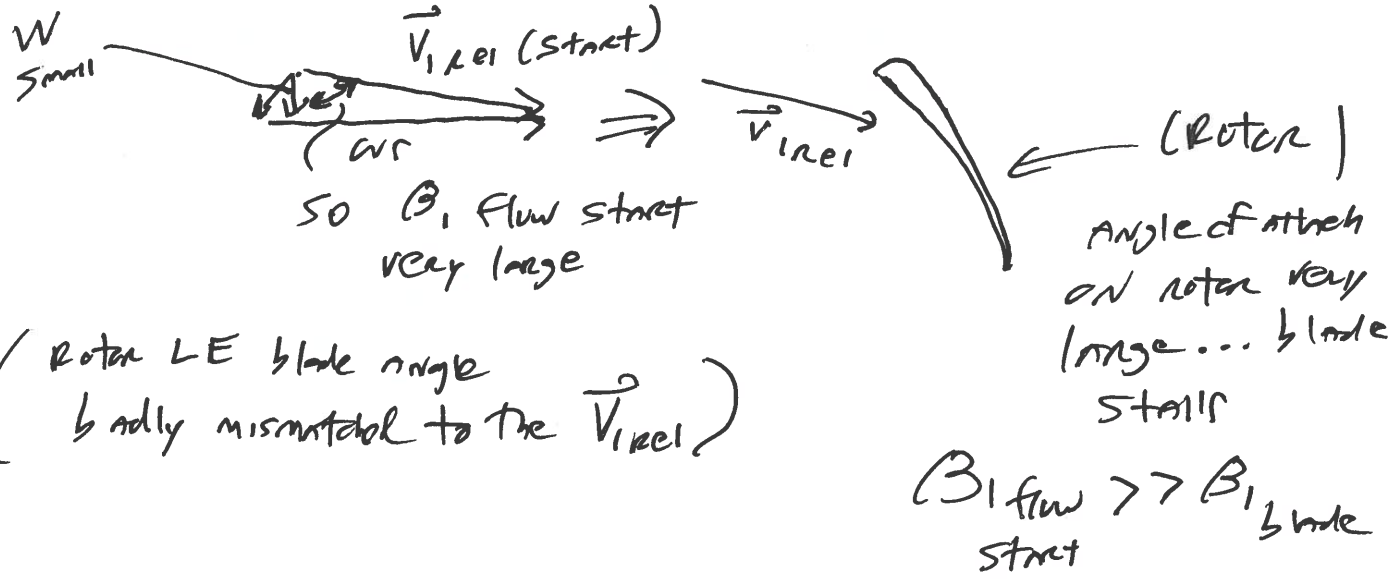
"Normal" Flow: (Blade angles, W, ωr 'matched') such that $\beta_{1\,flow} \sim \beta_{1\,blade\ angle}$



So $\beta_{1\,flow} \sim \beta_{1\,blade}$

Angle of attack on Rotor blade small, little risk of separation

But when compressor is starting, W is small on initial stages. So → if stator & blade angles are not changed from 'normal run':



So, during starting, rotate upstream stator (clockwise in sketch) to make

$\beta_{1\,flow} \sim \beta_{1\,blade}$ starting

(upstream) Stator orientation:



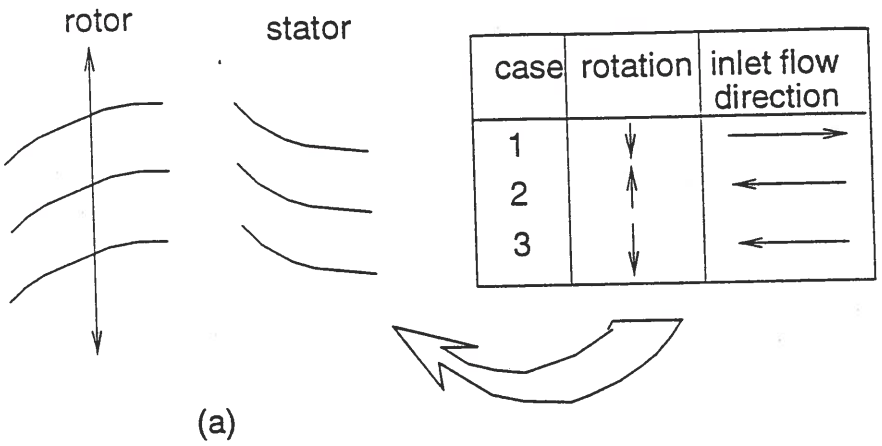
For rear stages, since overall area construction thru a compressor is based on 'design' ('run') conditions (to maintain axial velocities thru compressor), when compressor is starting, power into flow does not match the area contraction so axial flow tends to speed up

... speed up too much on rear stages; rear stages can 'wind-mill' (develop negative angles of attack - trying to act like a turbine)

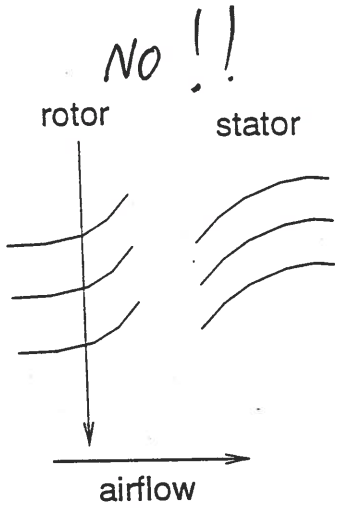
So for starting (& 'off-design operation), ECS must schedule both stator orientation and axial bleed to ensure effective operation. Also 'spooling' used (compressor-turbine are 'segmented' and run sections at different RPM) ...

Turbine aerodynamics analyzed in similar fashion. The primary function of the stator in turbine is to provide a 'high' velocity impinging on downstream rotor (to provide adequate force & have torque on rotor) ...

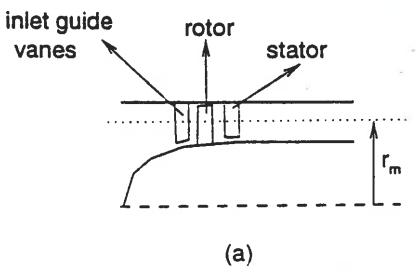
1. Shown schematically in part a) of the figure is the blading of a single-stage axial turbomachine. What kind of machine is represented by cases 1 and 2? What would happen in case 3? Would it be desirable to build a compressor stage as in part b) of the figure?

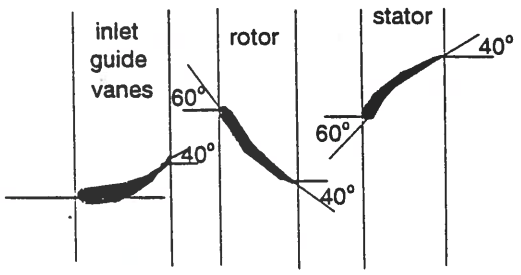


see next page



2. Estimate the power required to drive a single stage compressor shown schematically as in parts a) and b) of the sketch below. At the mean radius r(mean) = .3048m. The blade configuration at this mean radius is as shown in part b). For simplicity it is assumed that the air and the blade angles are identical. The overall efficiency of the stage is .8. The hub-tip radius ratio is 0.8 and is high enough so that conditions at the mass-averaged radius are a good average of conditions from hub to tip. Axial velocity component at design flow rate is uniformly 122 m/s and the inlet air is at 1 atm and 288K. What should the shaft speed be under these conditions? What is the ΔP_t through the stage?



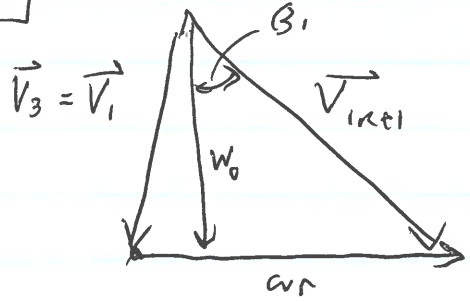


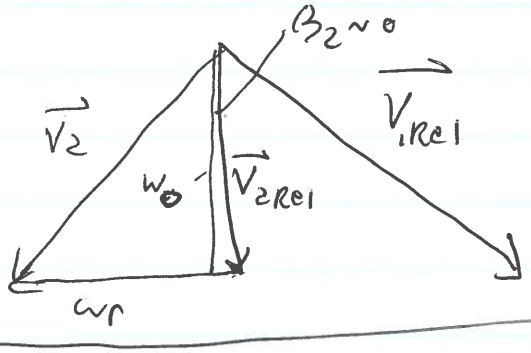
following pages ...

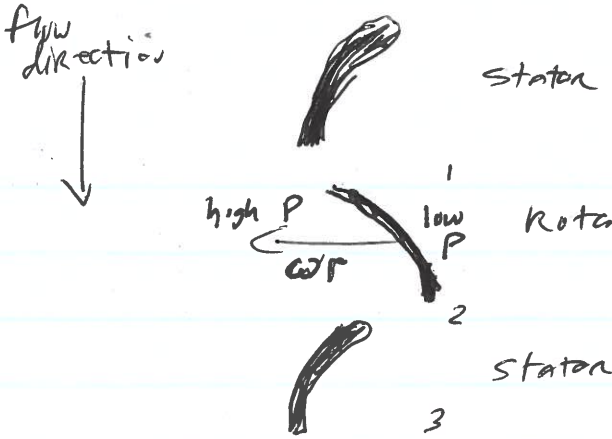
$\uparrow \omega r$

#1

CASE 1 :



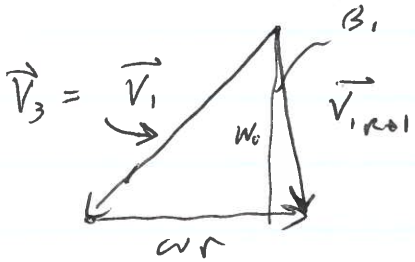


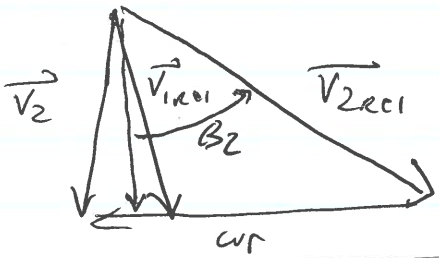


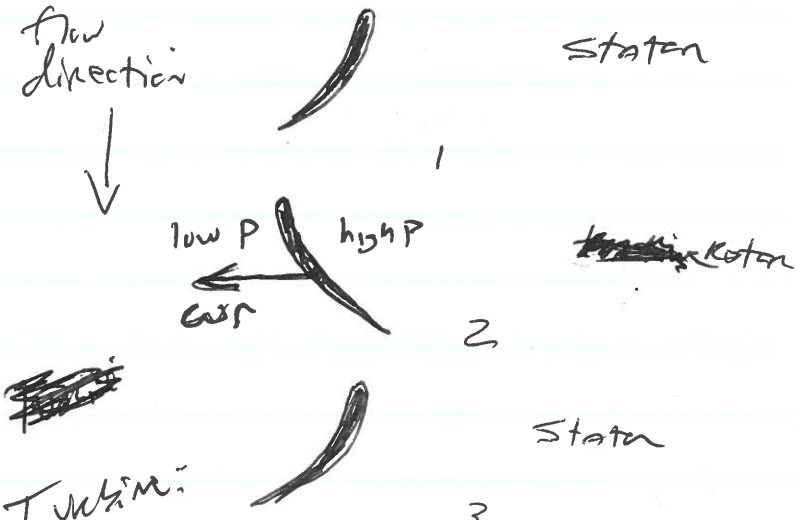
Compression : Rotor works on the flow, stator turns flow back toward Axial direction

Note: Rotor "pushing" against higher P side of Rotor blade - takes work !

CASE 2 :

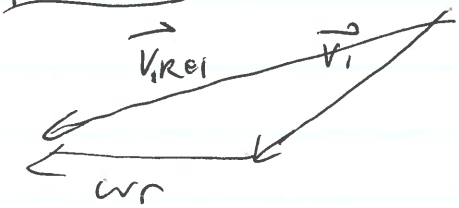


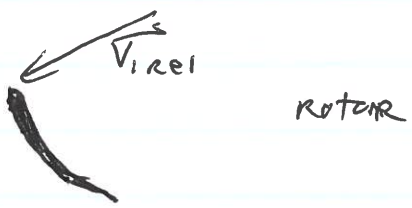




Note: Rotor being pushed by high P side - produces work !

CASE 3 (cwr changes sign)





MASSIVE mismatch between blade angle & flow direction

MASSIVE Stall

Solution: (#2)

$P_0 = 101325\ N/m^2$

$T_0 = 288\ K$

$(\rho_0 = 1.225\ kg/m^3)$

$W_0 = 122\ m/s$

$\eta_c = 0.8$

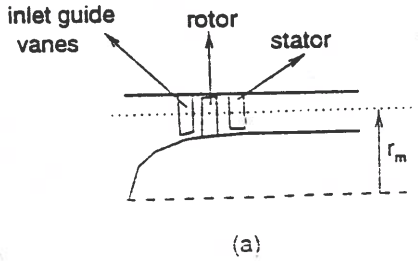
$\frac{r_t}{r_h} = \frac{1}{.8} = 1.25$

r_m = MASS AVERAGED RADIUS $= 0.3048\ m$

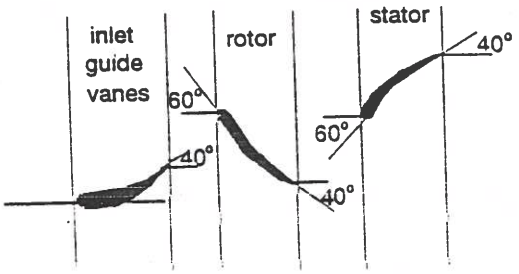
$M_0 = \frac{W_0}{\sqrt{\gamma R T_0}} = 0.359$

$T_{t0} = 295.41\ K$

$P_{t0} = 110745\ N/m^2$



(a)



(b)

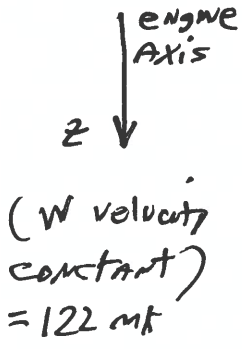
z
engine Axis (main flow direction)

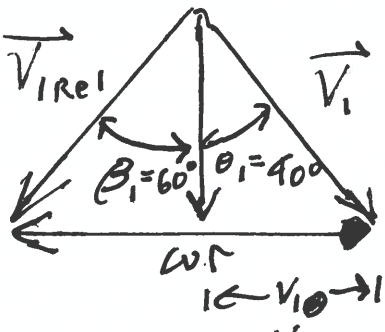
ωr (Rotor Rotation direction)

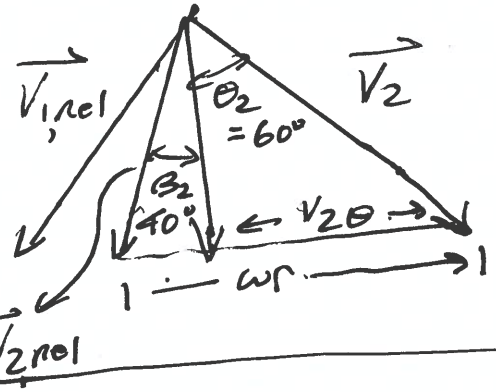
$$\text{Power required} = \omega\left[(cr V_\theta)_2 - (cr V_\theta)_1\right]\dot{m}$$

0-1 IGV
1-2 Rotor
2-3 stator
1

Velocity triangles:







$W_0 = W_1 = W_2$

$\theta_1 = 40°$ $\quad \theta_2 = 60°$

$\beta_1 = 60°$ $\quad \beta_2 = 40°$

$V_{1\theta} = 102.4\ m/s$

$V_{2\theta} = 211.3\ m/s$

$$\left\{ \frac{\omega r - V_{\theta 1}}{W_0} = \tan \beta_1 \qquad \frac{\omega r - V_{\theta 2}}{W_0} = \tan \beta_2 \right\}$$

$$\left\{ \frac{V_{\theta 1}}{W_0} = \tan \theta_1 \qquad \frac{V_{\theta 2}}{W_0} = \tan \theta_2 \right\}$$

so $\frac{\omega r}{W_0} = 2.57 \Rightarrow$ **$\omega = 1029$ rad/sec ($= 9826$ RPM)**

Need mi thru stage (for ✱):

$r_m = 0.30$ ft ← mass average radius

if W uniform with r ...

$$\frac{r_t^2 + r_h^2}{2} = r_m^2$$

& since $\frac{r_t}{r_h} = 1.25$ (given)

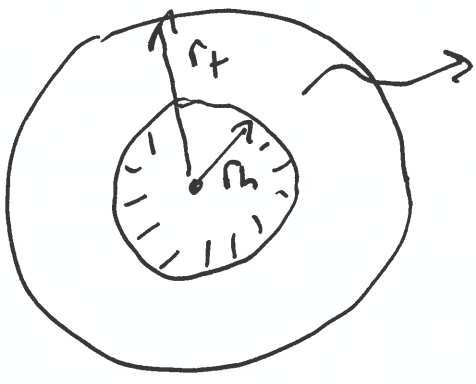
so $\left\{ \begin{array}{l} r_h = 0.2693 \text{ m} \\ r_t = 0.3366 \text{ m} \end{array} \right\}$

so A (cross-sectional area of flow path) $= A_{tip} - A_{hub}$

$$= \pi r_t^2 - \pi r_h^2$$

$A = 0.128 \text{ m}^2$

$\dot{m} = \rho_0 W_0 A = 19.16$ kg/sec



assuming adiabatic blades!

from * $\text{Power} = \dot{m}(h_{t_2} - h_{t_1}) = \dot{m} c_w \left[(rV_\theta)_2 - (rV_\theta)_1\right]$

$\left(= \dot{m} c_p (T_{t_2} - T_{t_1})\right)$

Power = 650,416 Watts ((Power necessary for this stage))

(also $T_{t_2} = 329.2\ K$)

How do we find ΔP_t (across stage) — Not isentropic → $\eta < 1$ given!!

let $\tau = \frac{T_{t_2}}{T_{t_0}}$ (Recall $T_{t_2} = 329.2\ K$)

stage based → $\pi_c = \frac{P_{t_2}}{P_{t_0}}$ (& $P_{t_0} = 110745\ N/m^2$)

& by definition of η_{stage}

$$\tau = 1 + \frac{\pi^{\frac{\gamma-1}{\gamma}} - 1}{\eta_c} \Rightarrow \pi = 1.3582$$

(across stage)

so $P_{t_2} = 150412\ N/m^2$

& $\Delta P_{t\{\text{actual} \to \text{for given } \eta\}} = 39669\ N/m^2$

Now, let us compare that to an isentropic stage & simply use blade angles info to get ΔP_t ...

$$** \quad \frac{\Delta P_t (\text{isentropic})}{\rho_0 W_0^2} = \frac{\sigma r}{W_0} (\tan\beta_1 - \tan\beta_2)$$

→ (Recall that we derived ** based on isentropic flow thru IGV, Rotor, stator; AND small M_0 → BIG ASSUMPTION AND small flow turning ...

so ... $\Delta P_t (\text{isentropic}) = 41873 \text{ N/m}^2$

(Compared to ΔP_t actual = 39669 N/m²)

... But the assumption that M_0 is small leads to a pretty large approximation (in the development of **) & hence in the ΔP_t (isentropic) ** equation ...

take ** AS approximate only

** is really strictly valid for incompressible ($\rho \sim \rho_{TOTAL}$) ($M \to 0$) & small turning ...
Note $\Delta\beta$ thru Rotor is ~20° (Not really small)

Turbine Aerodynamics

* Usually in turbines, pressure drops across individual stages are so great that compressibility of flow becomes significant in design/analysis. This is due to the fact that turbines have favorable pressure gradients and hence can support large pressure changes..
* In general, the stator in the turbine changes the direction of the flow (it introduces 'swirl' component to the velocity) with an associated static pressure drop (i.e., velocity increases). The turbine rotor removes some portion of the kinetic energy identified with the swirl. The axial velocity through the stage is kept fairly constant (at least near the acceptable design/off-design point range) by incorporating an <u>area increase</u> in the overall flow.
* The stator is called the 'turbine nozzle' due to its characteristic of increasing the flow velocity

---

* <u>Impulse stage</u> (or 0° Reaction stage)

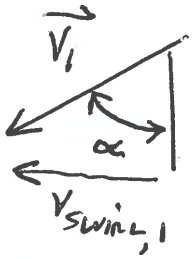
{ Recall $0R = \frac{\Delta P_{Rotor}}{\Delta P_{stage}}$ (degree of reaction) so ΔP's negative for turbine stage) }

A turbine stage in which the <u>entire</u> static pressure drop occurs solely in the stator is an <u>impulse stage</u>. This implies that the relative velocity of the flow thru the rotor does not change magnitude.
If rotor leaves <u>NO</u> residual swirl, this is a <u>maximum work turbine stage</u>.

Impulse Turbine Stage (0° Reaction)

$|\vec{V}_2| \sim W_0$ (from previous stage)

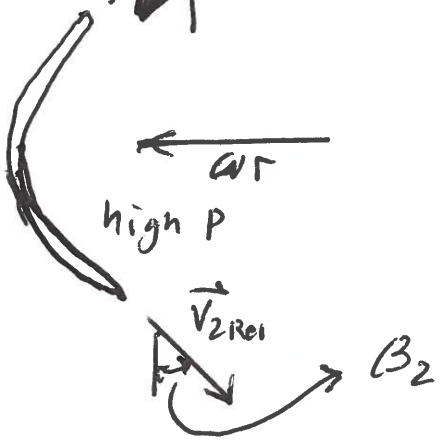
[STATOR FRAME]



* Note correspondence of α to compression 'θ'
* α = 'SWIRL ANGLE'
* $V_{SWIRL_1} = V_1 \sin\alpha$ = swirl velocity component introduced by stator

(1)



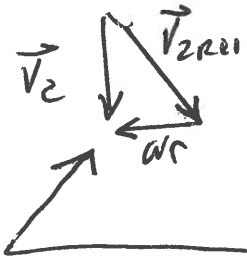


[ROTOR FRAME]

SINCE $\frac{\Delta P_{Rotor}}{\Delta P_{Stage}} \sim 0$;

$|\vec{V}_{1REL}| \sim |\vec{V}_{2REL}|$

(2)



$\vec{V}_2 \sim W_0$

(NO SWIRL velocity left after rotor

$V_{SWIRL,2} = 0$

NEXT STATOR FRAME

IF NO Residual SWIRL velocity after Rotor → 'MAXIMUM WORK TURBINE'

* 50% Reaction Stage

(see cascade view next page)

A turbine stage which has equal static pressure drops for both stator AND Rotor would be a 50% Reaction stage (Hence $^{\circ}R = 50\%$)

* Can have 30% Reaction stages, etc...

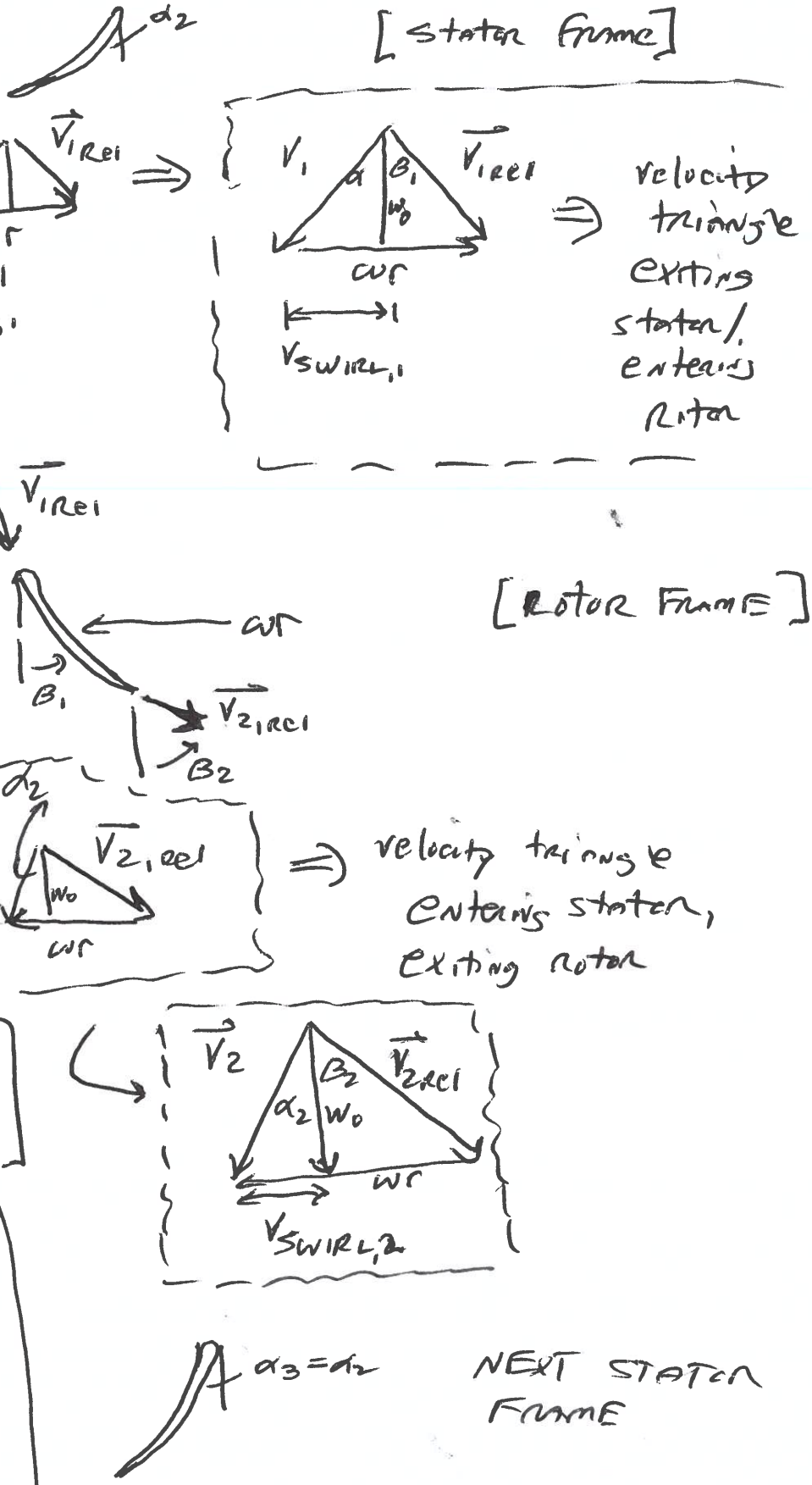
* Impulse stages can be shown to yield maximum work interaction (^{MAX} power provided) due to the fact that greater swirl velocity is introduced to the rotor by the stator, and 'entirely' removed by the rotor.

* A 50% Reaction stage is better from an efficiency standpoint, but produce less work per stage. Hence, in practice, impulse turbines are often used in high thrust-to-weight engines due to very large power capability per stage. Transport aircraft in which design is usually concerned with efficiency usually uses 30-50% Reaction turbine stages. Tradeoff is obviously number of stages / weight...

50% Reaction Stage

(Turbine)

$\vec{V}_2$ (from previous stage)



Note $V_{SWIRL,2} \neq 0$ (but $< V_{SWIRL,1}$)

Rotor does not remove all of swirl velocity in 50% reaction stage

Relative Stagnation temperatures in turbines

Due to the fact that the turbine rotor blades are turning in the direction of the flow (i.e., the flow is working on the blades and 'pushing' them), the relative flow velocities seen by the turbine blades are somewhat reduced (opposite in a compressor). [But compressor generally operates with lower flow velocities due to adverse pressure gradients than turbine]. Hence stagnation temperatures on turbine blades are somewhat reduced. Keep in mind, though, that turbine blades (particularly the first stages) still exist in a more severe thermal environment due to the upstream burner!

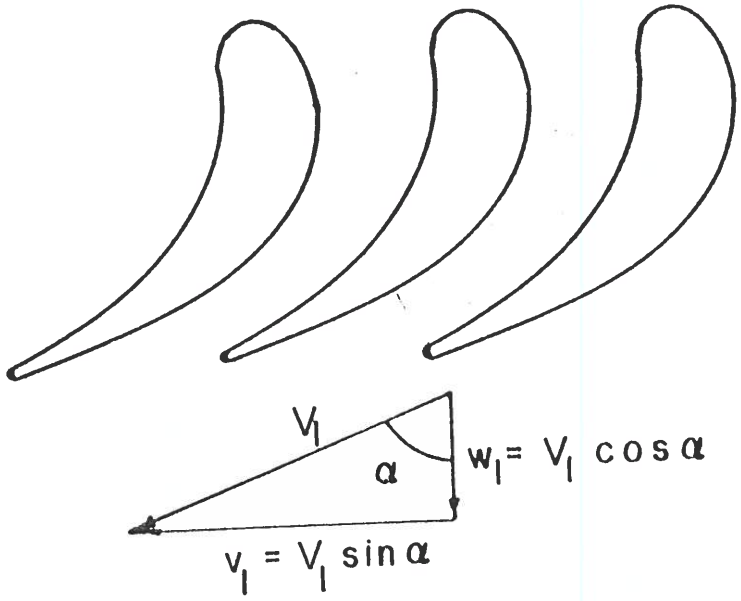
Let T_{t_R} = 'relative' stagnation temperature on blade surfaces:

Analysis shows...

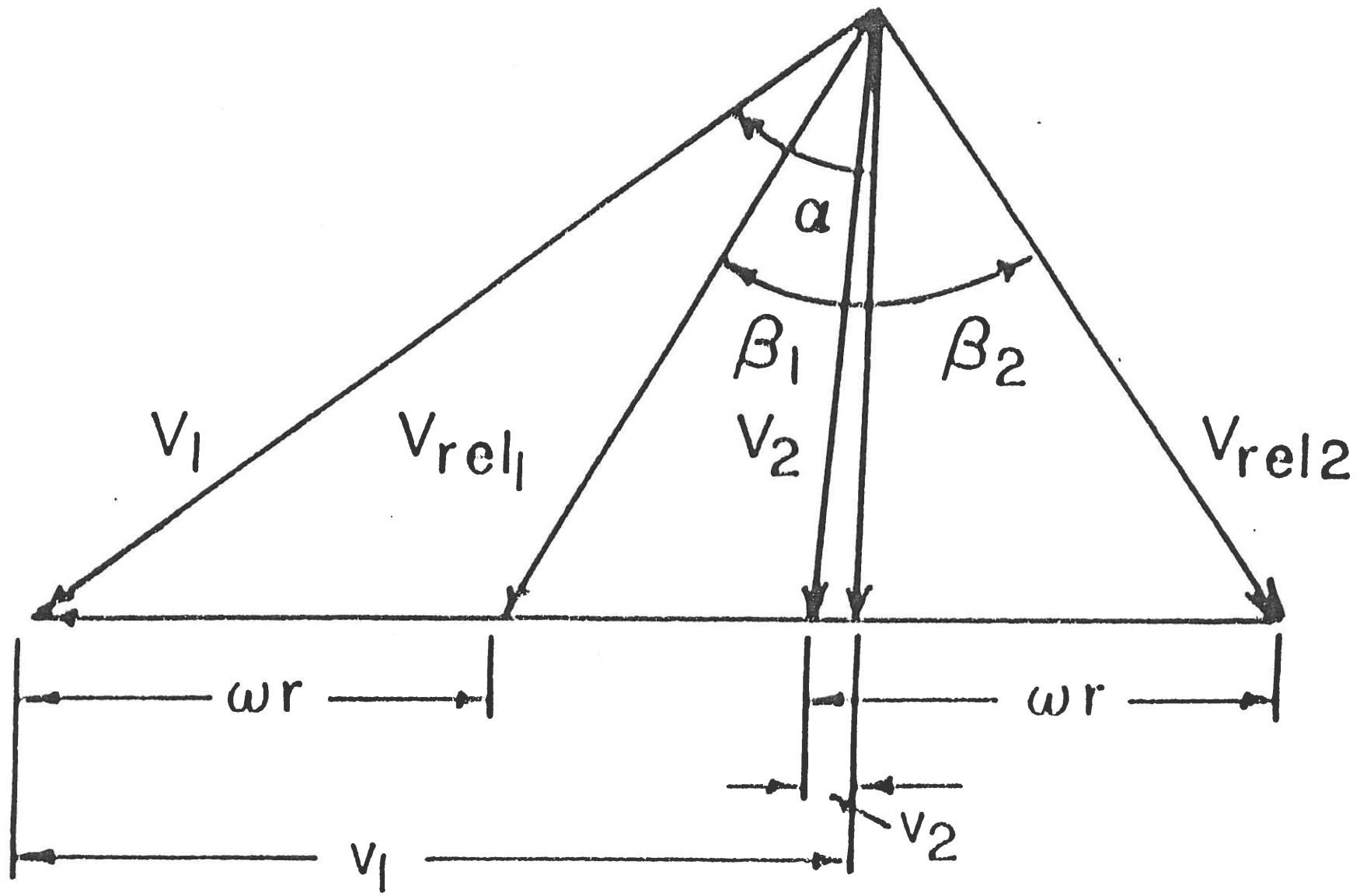
$$\frac{T_{t_R}}{T_{t_1}} = 1 - \frac{3/2\,(\omega r)^2}{2 C_p T_{t_1}}$$ Maximum work impulse turbine

$$\frac{T_{t_R}}{T_{t_1}} = 1 - \frac{1/2\,(\omega r)^2}{2 C_p T_{t_1}}$$ 50% reaction turbine

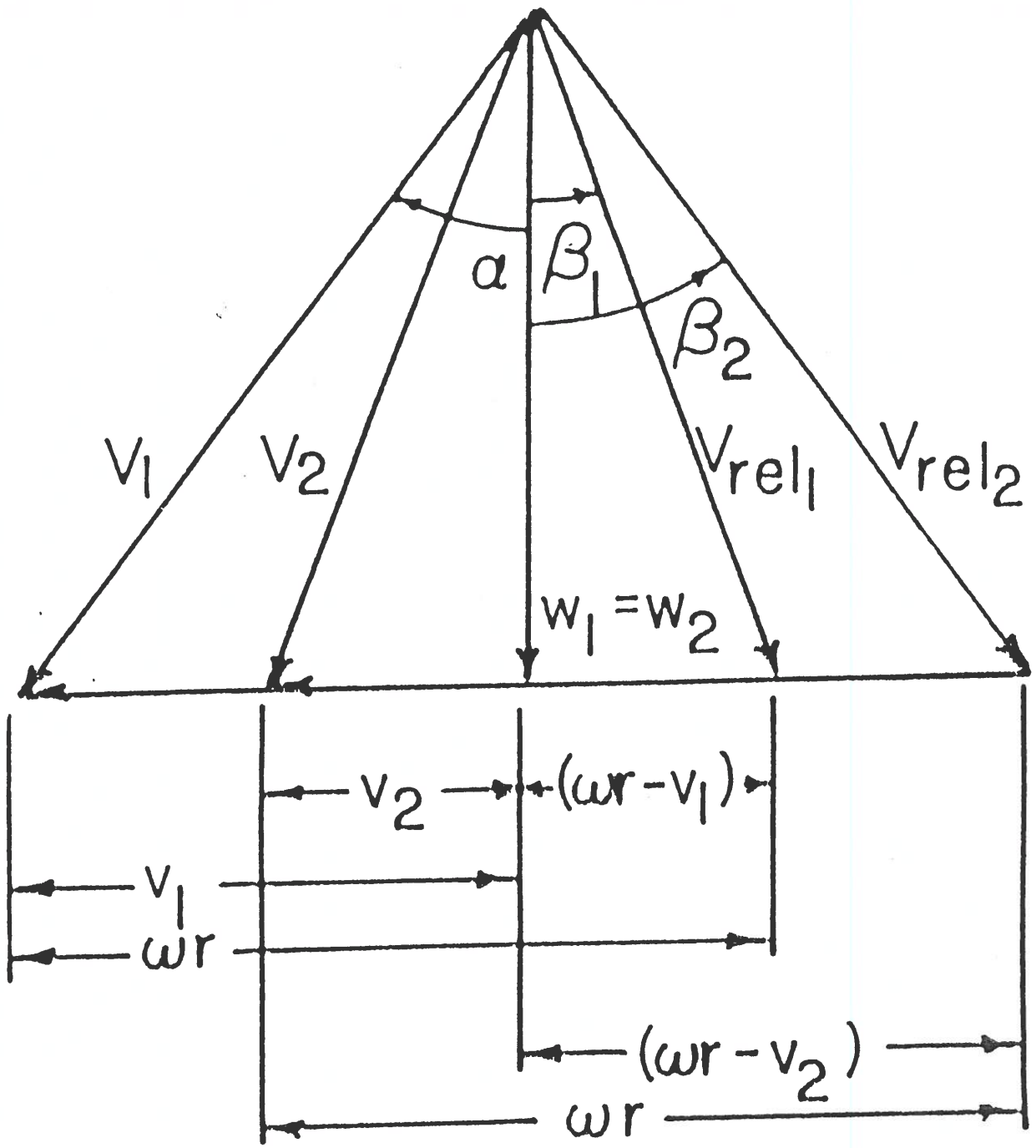
[Hence impulse turbines 'see' lower stagnation temperatures than 50% reaction turbines]



Turbine nozzle row. (Stator)



Velocity diagram for an impulse turbine stage.



Velocity diagram for a reaction turbine stage.